



I will give you some pictures from notes of probability and statistics. You will explain those very very clearly with some intuition to make me understand things better and make my fuully examprepared. got it?

Got it 

Please send me the pictures of your probability and statistics notes. I'll carefully explain each one in very clear detail, step by step, with intuition and examples so you'll be fully prepared for your exam.

Would you like me to also add **short formulas + exam tricks/tips** along with the explanations?



Perfect 🧙 You've given me your **Probability & Statistics notes on Covariance and Correlation**.

Let me explain everything step by step with **intuition + formulas + exam-friendly points**.

◆ 1. Why Covariance and Correlation?

- When we have **two random variables** (say height & weight, or study hours & exam score), we want to know **how they are related**.
 - **Covariance & Correlation** are **numerical measures of relationship (association)**.
 - **Covariance**: Measures whether they move **together** (positive), **opposite** (negative), or are **unrelated** (zero).
 - **Correlation**: Scales this relationship between **-1 and +1** for easier interpretation.
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◆ 2. Definition of Covariance

Your notes define it as:

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

👉 Meaning:

Take each variable's **deviation from its mean**, multiply them, then take the expectation (average).

♦ Intuition

- If **X is above its mean** and **Y is also above its mean**, the product $(X - \mu_X)(Y - \mu_Y)$ is **positive**.
- If **X is above mean** but **Y is below mean**, the product is **negative**.
- Covariance averages all these products.

So:

- $\text{Cov}(X, Y) > 0 \rightarrow$ X and Y tend to **increase together** (positive relationship).
 - $\text{Cov}(X, Y) < 0 \rightarrow$ X increases while Y decreases (negative relationship).
 - $\text{Cov}(X, Y) = 0 \rightarrow$ No linear relationship.
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♦ 3. Simplified Formula

Expanding:

$$\text{Cov}(X, Y) = E[XY] - \mu_X \mu_Y$$

👉 Much easier for calculation:

1. Find $E[XY]$ (expectation of product).
 2. Find product of means $\mu_X \mu_Y$.
 3. Subtract.
-

♦ 4. Independence Case

If X and Y are **independent**:

$$E[XY] = E[X] \cdot E[Y]$$

So:

$$\text{Cov}(X, Y) = E[X]E[Y] - \mu_X\mu_Y = 0$$

👉 Independence $\Rightarrow$ Covariance = 0.

(but converse is not always true: Cov=0 does **not always mean independence**).

♦ 5. Example (for intuition)

Imagine:

- X : Hours studied.
- Y : Exam score.

Case 1 (positive covariance):

More study hours $\rightarrow$ higher exam scores.

$$\text{Cov}(X, Y) > 0.$$

Case 2 (negative covariance):

More time on mobile $\rightarrow$ lower exam scores.

$$\text{Cov}(X, Y) < 0.$$

Case 3 (zero covariance):

Shoe size vs. exam score.

$$\text{No relationship} \rightarrow \text{Cov}(X, Y) = 0.$$

✅ Exam Tip:

- Always remember the short formula:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

- Independence $\Rightarrow$ Cov=0.
- But Cov=0 does not guarantee independence.

Would you like me to also cover **Correlation** (since your notes mention it next) with clear intuition and formula, so you have the full picture?



Let's break down the **Properties of Covariance** in the next few steps with **clear intuition** and **proofs**.

1. Property (1):

$$\text{Cov}(X, X) = \text{Var}(X)$$

- **Intuition:** The covariance of a random variable with itself is simply its **variance**.
 - **Why?:** The variance is just the expectation of the square of the deviations from the mean, which is exactly how covariance with itself is defined.

Proof (Property 1):

- By definition of covariance:

$$\text{Cov}(X, X) = E[(X - \mu_X)(X - \mu_X)] = E[(X - \mu_X)^2]$$

This is exactly the definition of the **variance**, so:

$$\text{Cov}(X, X) = \text{Var}(X)$$

2. Property (2):

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

- **Intuition:** Covariance is **symmetric**, meaning the relationship between **X** and **Y** is the same as the relationship between **Y** and **X**.
- **Why?:** The definition involves a product of deviations $(X - \mu_X)(Y - \mu_Y)$, and multiplication is commutative, so it doesn't matter in which order you multiply X and Y .

Proof (Property 2):

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[(Y - \mu_Y)(X - \mu_X)] = \text{Cov}(Y, X)$$

3. Property (3):

$$\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$$

- **Intuition:** If you scale both variables, the covariance scales by the product of the scales.
- **Why?:** Multiplying X by a and Y by b stretches the deviations by those amounts, so the covariance will also scale by ab .

Proof (Property 3):

$$\text{Cov}(aX, bY) = E[(aX - a\mu_X)(bY - b\mu_Y)] = ab E[(X - \mu_X)(Y - \mu_Y)] = ab \text{Cov}(X, Y)$$

4. Property (4):

$$\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$$

- **Intuition:** If you combine two variables (say X and Y) and look at the covariance with another variable Z , the total covariance is the sum of the individual covariances.
- **Why?:** This is because covariance is linear with respect to both terms. Each term can be expanded separately and then summed.

Proof (Property 4):

$$\begin{aligned}\text{Cov}(X + Y, Z) &= E[(X + Y - \mu_{X+Y})(Z - \mu_Z)] \\ &= E[(X - \mu_X + Y - \mu_Y)(Z - \mu_Z)]\end{aligned}$$

Expanding and simplifying:

$$= E[(X - \mu_X)(Z - \mu_Z)] + E[(Y - \mu_Y)(Z - \mu_Z)] = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$$

Quick Recap & Exam Tips:

- **Covariance with itself → Variance:** $\text{Cov}(X, X) = \text{Var}(X)$
- **Symmetry:** $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
- **Scaling:** $\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$
- **Addition:** $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$

Would you like to dive deeper into **correlation** (which is the next logical step) or need any further explanation on **covariance** properties?



Let's continue with the **variance of the sum of random variables**.

Variance of the Sum of Random Variables

The formula from your notes is:

$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$$

This is a key property when dealing with the **variance of the sum** of multiple random variables.

Intuition Behind the Formula:

- The total **variance** of the sum of random variables is the sum of individual variances, **plus** twice the sum of covariances between pairs of random variables.
- **Why?**
 - Each random variable X_i contributes to the total variance.
 - The **covariances** measure the relationship between pairs of variables. If two variables are **related**, their covariance is non-zero, and it adds to the

total variance.

- The **factor of 2** comes from summing over all pairs (i, j) , where each pair will contribute twice to the total.

Proof (Breakdown)

Starting with the definition of **variance**:

$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \text{Cov} \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i \right)$$

By expanding this:

$$= \sum_{i=1}^n \text{Cov}(X_i, X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$$

The first part is simply:

$$\sum_{i=1}^n \text{Cov}(X_i, X_i) = \sum_{i=1}^n \text{Var}(X_i)$$

And the second part is the **sum of covariances** between distinct pairs of variables:

$$2 \sum_{i < j} \text{Cov}(X_i, X_j)$$

Special Case: When Variables Are Independent

If the random variables $X_1, X_2, \dots, X_n$ are **independent**, then:

$$\text{Cov}(X_i, X_j) = 0 \quad \text{for all } i \neq j$$

Thus, the formula simplifies to:

$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i)$$

This means that when the variables are **independent**, the **total variance** is simply the sum of the individual variances (without any need for covariances).

Quick Recap & Exam Tips:

- **Variance of Sum:**

$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$$

- If the variables are **independent**, the covariances vanish, and the formula simplifies to just the sum of individual variances.

Would you like further elaboration on **covariance** or would you like to continue with **correlation**?



explain this very very clearly

Great—this page is the classic result for the **variance of a sum**. I'll explain it slowly, give intuition, show the algebra, and point out exam traps.

Theorem (what it says)

For random variables $X_1, \dots, X_n$ with finite variances,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$$

- The first term is the sum of individual spreads.
- The second term adds the **pairwise relationships** (co-movements).
Positive covariance **increases** the total spread; negative covariance **reduces** it.

If the X_i are **independent**, then every $\text{Cov}(X_i, X_j) = 0$ for $i \neq j$, so

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i).$$

Intuition (why it's true)

- Variance measures how much a total fluctuates around its mean.
- If two variables **tend to move together** (positive covariance), their ups and downs **reinforce**, making the sum more volatile $\Rightarrow$ add positive amount.
- If they **counter-move** (negative covariance), they **cancel** fluctuations $\Rightarrow$ subtract amount.
- If they're **unrelated** (covariance = 0), their co-movements don't systematically affect the sum.

Proof (line by line, matching your notes)

Let $S = \sum_{i=1}^n X_i$.

1) Connect variance and covariance

$$\text{Var}(S) = \text{Cov}(S, S).$$

2) Use bilinearity (additivity) of covariance

For any A, B, C : $\text{Cov}(A + B, C) = \text{Cov}(A, C) + \text{Cov}(B, C)$, and similarly in the

second slot.

$$\text{Cov}\left(\sum_{i=1}^n X_i, \sum_{j=1}^n X_j\right) = \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j).$$

3) Split the double sum into diagonal and off-diagonal parts

$$\sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j) = \underbrace{\sum_{i=1}^n \text{Cov}(X_i, X_i)}_{\text{diagonal}} + \underbrace{\sum_{i \neq j} \text{Cov}(X_i, X_j)}_{\text{off-diagonal}}.$$

4) Identify variance on the diagonal

$$\text{Cov}(X_i, X_i) = \text{Var}(X_i) \Rightarrow \sum_{i=1}^n \text{Cov}(X_i, X_i) = \sum_{i=1}^n \text{Var}(X_i).$$

5) Handle off-diagonal terms

Each unordered pair $\{i, j\}$ appears **twice** in the double sum: (i, j) and (j, i) , and $\text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i)$.

So

$$\sum_{i \neq j} \text{Cov}(X_i, X_j) = 2 \sum_{i < j} \text{Cov}(X_i, X_j).$$

Combine steps 4 and 5 to get the boxed formula.

Small-n snapshots (to see the structure)

- $n = 2$:

$$\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2) + 2 \text{Cov}(X_1, X_2).$$

- $n = 3$:

$$\text{Var}(X_1 + X_2 + X_3) = \sum_{i=1}^3 \text{Var}(X_i) + 2[\text{Cov}(X_1, X_2) + \text{Cov}(X_1, X_3) + \text{Cov}(X_2, X_3)].$$

Quick computational recipe (exam-friendly)

1. Compute each $\sigma_i^2 = \text{Var}(X_i)$.
2. For each pair $i < j$, compute $\text{Cov}(X_i, X_j)$.
If you're given correlations ρ_{ij} : $\text{Cov}(X_i, X_j) = \rho_{ij}\sigma_i\sigma_j$.
3. Plug into

$$\sum \sigma_i^2 + 2 \sum_{i < j} \text{Cov}(X_i, X_j).$$

Common pitfalls

- **Forgetting the factor 2** in the covariance sum. (It comes from counting both (i, j) and (j, i) .)
- **Assuming zero covariance $\Rightarrow$ independence.** Not necessarily; zero covariance only means **no linear** relationship.
- **Pairwise independence is enough** to drop the covariance terms; you don't need full mutual independence.

If you want, send a small numeric problem (variances + correlations), and I'll compute $\text{Var}(\text{sum})$ step-by-step to lock it in.



can u suggest and youtube video that explains this?

Here's an excellent video that walks through the **variance of the sum of random variables**, with clear explanation and examples:

An error occurred.

Try watching this video on www.youtube.com, or enable JavaScript if it is disabled in your browser.

Why This Video Is Especially Helpful:

- Direct explanation of the key formula:

$$\text{Var}\left(\sum X_i\right) = \sum \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$$

- Step-by-step breakdown with intuitive reasoning and practical context.
- Great pacing—easy to follow for anyone aiming to solidify their understanding.

Bonus Options You Might Like:

- **MIT OpenCourseWare - “L12.7 The Variance of the Sum of Random Variables”**

A lecture-style explanation from MIT’s probability course. Great for a deeper, classroom-style walkthrough.

[youtube.com+1math.stackexchange.com+1probabilitycourse.com](#)
[youtube.com+2](#)

- **Khan Academy - “Variance of sum and difference of random variables”**

Friendly, approachable, with intuitive visuals and helpful for reinforcing the concept.

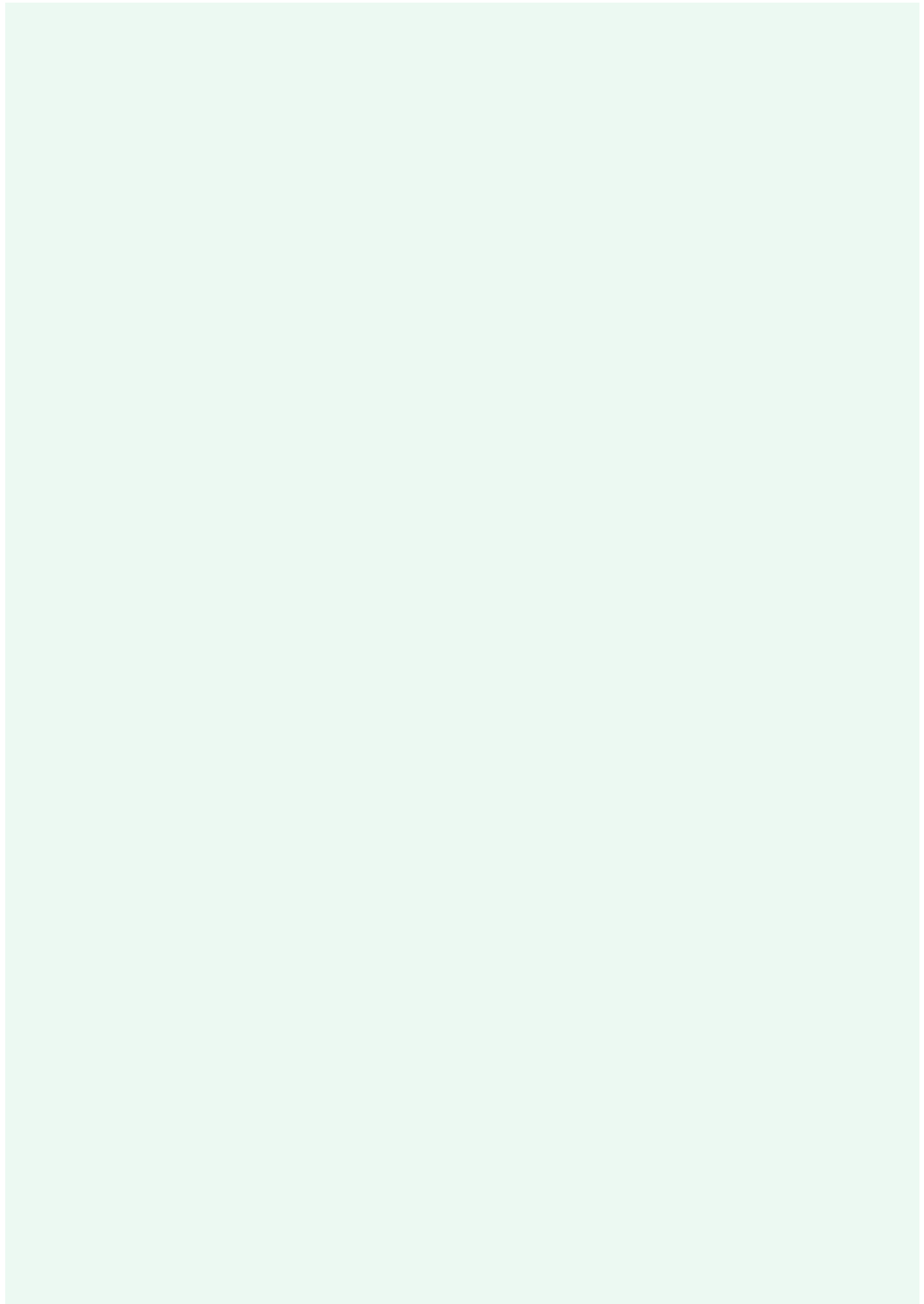
[khanacademy.org+2](#)

How to Use These Videos Effectively:

1. **Watch** and pause when key steps are shown.
2. **Take notes**—especially on how they derive the formula and handle covariance.
3. **Recreate the derivation yourself** on paper while watching.
4. **Try your own examples**, perhaps with two or three variables, including cases with and without covariance.
5. **Teach it to someone else** or explain aloud—you’ll remember it far better.

Need help walking through how any of those examples work, or want a short quiz-style check to make sure you're ready? Just ask!





explain this step by step very clearly

Awesome—this is a full worked example of finding $\text{Cov}(\mathbf{X}, \mathbf{Y})$ from a joint pdf. I'll go step-by-step and explain *why* each step works.

Given (joint pdf)

$$f_{X,Y}(x, y) = \begin{cases} \frac{6}{5} (x^2 + 2xy), & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

We want $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$.

0) Quick validity check (always good to do)

A joint pdf must integrate to 1:

$$\iint_{[0,1]^2} \frac{6}{5} (x^2 + 2xy) dx dy = \frac{6}{5} \left(\underbrace{\int_0^1 x^2 dx}_{1/3} \underbrace{\int_0^1 dy}_1 + 2 \underbrace{\int_0^1 x dx}_{1/2} \underbrace{\int_0^1 y dy}_{1/2} \right) = \frac{6}{5} \left(\frac{1}{3} + \frac{1}{2} \right) = 1$$

So it's a valid pdf.

1) Marginal pdfs

Marginal of X : $f_X(x) = \int_0^1 f_{X,Y}(x, y) dy$

$$f_X(x) = \frac{6}{5} \int_0^1 (x^2 + 2xy) dy = \frac{6}{5} \left(x^2 \cdot 1 + 2x \cdot \frac{1}{2} \right) = \frac{6}{5} x(x + 1), \quad 0 \leq x \leq 1.$$

Marginal of Y : $f_Y(y) = \int_0^1 f_{X,Y}(x, y) dx$

$$f_Y(y) = \frac{6}{5} \int_0^1 (x^2 + 2xy) dx = \frac{6}{5} \left(\frac{1}{3} + y \right) = \frac{2}{5} (1 + 3y), \quad 0 \leq y \leq 1.$$

(Tip: we can factor constants because the region is a rectangle $[0, 1] \times [0, 1]$.)

2) Means $E[X]$ and $E[Y]$

$$E[X] = \int_0^1 x f_X(x) dx = \int_0^1 x \cdot \frac{6}{5} x(x+1) dx = \frac{6}{5} \int_0^1 (x^3 + x^2) dx = \frac{6}{5} \left(\frac{1}{4} + \frac{1}{3} \right)$$

$$E[Y] = \int_0^1 y f_Y(y) dy = \int_0^1 y \cdot \frac{2}{5} (1 + 3y) dy = \frac{2}{5} \int_0^1 (y + 3y^2) dy = \frac{2}{5} \left(\frac{1}{2} + 1 \right) = \frac{6}{5}$$

3) The mixed mean $E[XY]$

Use the joint pdf directly:

$$E[XY] = \iint_{[0,1]^2} xy f_{X,Y}(x,y) dx dy = \frac{6}{5} \iint_{[0,1]^2} (x^3 y + 2x^2 y^2) dx dy.$$

Because each term factors into an x -part times a y -part, split the integrals:

$$E[XY] = \frac{6}{5} \left(\underbrace{\int_0^1 x^3 dx}_{1/4} \underbrace{\int_0^1 y dy}_{1/2} + 2 \underbrace{\int_0^1 x^2 dx}_{1/3} \underbrace{\int_0^1 y^2 dy}_{1/3} \right) = \frac{6}{5} \left(\frac{1}{8} + \frac{2}{9} \right) = \frac{6}{5} \cdot \frac{25}{72} = \frac{5}{12}$$

4) Covariance

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = \frac{5}{12} - \left(\frac{7}{10} \right) \left(\frac{3}{5} \right) = \frac{5}{12} - \frac{21}{50} = \frac{125 - 126}{300}$$

Interpretation: Slight **negative** linear association (very small in magnitude).

5) Independence check (sanity)

If X, Y were independent, $f_{X,Y}(x, y) = f_X(x)f_Y(y)$.

Here,

$$f_X(x)f_Y(y) = \frac{6}{5}x(x+1) \cdot \frac{2}{5}(1+3y) = \frac{12}{25}x(x+1)(1+3y)$$

which is **not** equal to $\frac{6}{5}(x^2 + 2xy)$ in general $\Rightarrow$ not independent $\Rightarrow$ a nonzero covariance (indeed $-1/300$) makes sense.

Mini-checklist you can reuse

1. Verify pdf integrates to 1.
2. Get marginals f_X, f_Y .
3. Compute $E[X], E[Y]$.
4. Compute $E[XY]$ from the joint.
5. Use $\text{Cov} = E[XY] - E[X]E[Y]$.

Want me to also compute **Var(X), Var(Y), and the correlation** ρ_{XY} for this pdf to complete the picture?